

Homework #1

Assigned: Jan 29, 2026

Due: Feb 5, 2026

1) Given $c_1 = 2 + j$ and $c_2 = -2 + j3$, calculate the following:

(a) $c_1 + c_2$

(b) $c_1 - c_2$

(c) $c_1 c_2$

(d) c_1 / c_2

Give the answers in both rectangular and polar forms.

A) $(2 + j) + (-2 + j3) = 0 + 4j = 4e^{j90^\circ}$

B) $(2 + j) - (-2 + j3) = 4 - 2j = 4.47e^{-j26.6^\circ}$

C) $(2 + j)(-2 + j3) = -4 + 6j - 2j + 3j^2 = -7 + 4j = 8.06e^{j150.3^\circ}$

D) $\frac{2 + j}{-2 + j3} \times \frac{-2 - j3}{-2 - j3} = \frac{-4 + 3}{4 + 9} + j \frac{-2 - 6}{4 + 9} = -\frac{1}{13} - \frac{8}{13}j$

$\frac{c_1}{c_2} e^{j(\theta_1 + \theta_2)} = \frac{2.23}{3.6} e^{j(26.6 - 123.7)} = 0.62e^{-j97.1^\circ}$

2) See the three parts below. Give answers in rectangular form.

(a) Given the equation $z^2 = 1 + j$. Find two values of z that satisfy this equation.(b) Calculate $(4j)^{1/2}$.(c) Calculate $(1 + 2j)^3$.

A) $z^2 = 1 + j$ — convert to polar

$$|1 + j| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \tan^{-1}\left(\frac{1}{1}\right) = 45^\circ = \frac{\pi}{4}$$

$$1 + j = \sqrt{2} e^{j\pi/4}$$

$$(z^2 = r e^{j\theta}) \sqrt{\quad}$$

$$z = \sqrt{r} \cdot e^{j(\theta/2 + k\pi)}$$

$$= \sqrt{\sqrt{2}} \cdot e^{j(\frac{\pi}{8} + k\pi)} \quad k = \text{roots } (0, 1)$$

$$z = 2^{\frac{1}{2} \cdot \frac{1}{2}} \cdot e^{j(\frac{\pi}{8} + k\pi)}$$

$$z_0 = 2^{1/4} e^{j(\frac{\pi}{8})}; \quad z_1 = 2^{1/4} e^{j(\frac{\pi}{8} + \pi)}$$

$$z_1 = 1.10 + j0.46; \quad z_2 = -1.10 - j0.46$$

B) $(4j)^{1/2} = \sqrt{4j}$ ★

$$4j = 4(\cos(\frac{\pi}{2}) + j \sin(\frac{\pi}{2}))$$

$$|r| = 4$$

$$\theta = 90^\circ \text{ or } \frac{\pi}{2}$$

$$\sqrt{4j} = \sqrt{r} \cdot e^{j\frac{\theta}{2}}$$

$$= \sqrt{4} \cdot e^{j\frac{\pi}{4}}$$

$$= 2e^{j\frac{\pi}{4}}$$

$$= 2(\cos(\frac{\pi}{4}) + j \sin(\frac{\pi}{4}))$$

$$= 2\left(\frac{\sqrt{2}}{2} + j \frac{\sqrt{2}}{2}\right)$$

$$= \sqrt{2} + j\sqrt{2}$$

C) $(1 + 2j)^3$

$$(1 + 2j)(1 + 2j)(1 + 2j)$$

$$(1 + 2j + 2j + 4j^2)(1 + 2j)$$

$$(-3 + 4j)(1 + 2j)$$

$$-3 - 6j + 4j + 8j^2$$

$$-11 - 2j$$

3) Give the complex number that corresponds to the phasor representation of the following time-harmonic functions.

(a) $F(t) = 3 \sin(\omega t - \pi/2)$

(b) $G(t) = -2 \cos(\omega t)$

(c) $H(t) = 4 \cos(60\pi t - \pi/4)$

(d) $E(t) = 4 \sin \omega t + \cos \omega t$

A) $F(t) = 3 \sin(\omega t - \frac{\pi}{2})$ $\sin(x) = \cos(x - \frac{\pi}{2})$
 $= 3 \cos(\omega t - \pi)$ $-\frac{\pi}{2} - \frac{\pi}{2} = -\pi$
 $F(t) = 3 e^{-j\pi}$
 $= 3(\cos(\pi) + j \sin(\pi))$
 $= 3(1 + 0j)$
 $F(t) = 3$

B) $G(t) = -2 \cos(\omega t)$
 $-2 \cos(\omega t) = 2 \cos(\omega t + \pi)$
 $G(t) = 2 e^{j\pi} = 2(\cos(\pi) + j \sin(\pi))$
 $= -2 + j0$
 $G(t) = -2$

C) $H(t) = 4 \cos(60\pi t - \frac{\pi}{4})$
 $= 4 e^{-j\frac{\pi}{4}}$
 $= 4(\cos(\frac{\pi}{4}) + j \sin(\frac{\pi}{4}))$
 $= 4(\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2})$
 $H(t) = 2\sqrt{2} - j 2\sqrt{2}$

D) $E(t) = 4 \sin(\omega t) + \cos(\omega t)$
 $= 4 \cos(\omega t - \frac{\pi}{2}) + \cos(\omega t)$
 $= 4 e^{-j\pi/2}$
 $= 4(\cos(\frac{\pi}{2}) + j \sin(\frac{\pi}{2}))$
 $= 4(0 + j1)$
 $E(t) = -4j$

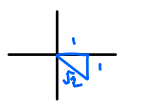
4) Obtain the sinusoidally-varying function $K(t)$ corresponding to the following phasors:

(a) $K = 1 - j$

(b) $K = e^{j0.4}$

(c) $K = 2e^{-j\pi/4} + 3e^{j0.3}$

A) $K = 1 - j$



$|r| = \sqrt{2}$
 $\theta = \tan^{-1}(\frac{-1}{1}) = -45^\circ = -\frac{\pi}{4}$
 $K = \sqrt{2} e^{-j\frac{\pi}{4}}$
 $K = \sqrt{2} \cos(\omega t - \frac{\pi}{4})$

B) $K = e^{j0.4}$

$|r| = 1$

$K = \cos(\omega t + 0.4 \text{ rads})$

C) $K = 2e^{-j\frac{\pi}{4}} + 3e^{j0.3}$

$2(\cos(\frac{\pi}{4}) + j \sin(\frac{\pi}{4}))$ $3(\cos(0.3) + j \sin(0.3))$
 $2(\frac{\sqrt{2}}{2} - j \frac{\sqrt{2}}{2})$ $3(1 + j0.0052)$
 $\sqrt{2} - j\sqrt{2}$ $(3 + j0.015)$

add

$4.41 - 1.4j$

$r = \sqrt{(4.41^2) + (1.4^2)}$ $\theta = \tan^{-1}(\frac{-1.4}{4.41})$
 $r = 4.63$ $\theta = -17.61^\circ$

$K(t) = 4.63 \cos(\omega t - 17.6^\circ)$

5) Let $\underline{A} = \hat{x} - 2\hat{y} + 3\hat{z}$ and $\underline{B} = -\hat{x} - 2\hat{y} + 3\hat{z}$

(a) Find $\underline{A} + \underline{B}$

(b) Find $\underline{A} - \underline{B}$

(c) Find $\underline{A} \cdot \underline{B}$

(d) Find $\underline{A} \times \underline{B}$

A) $\underline{A} + \underline{B} = (A_x + B_x) + (A_y + B_y) + (A_z + B_z)$
 $= (1 + (-1)) + (-2 + (-2)) + (3 + 3)$
 $= 0\hat{x} - 4\hat{y} + 6\hat{z}$

$\underline{A} + \underline{B} = -4\hat{y} + 6\hat{z}$

B) $\underline{A} - \underline{B} = (A_x - B_x) + (A_y - B_y) + (A_z - B_z)$
 $= (1 - (-1)) + (-2 - (-2)) + (3 - 3)$

$\underline{A} - \underline{B} = 2\hat{x} + 0\hat{y} + 0\hat{z}$

C) $\underline{A} = x - 2y + 3z$
 $\underline{B} = -x - 2y + 3z$

$\underline{A} \cdot \underline{B} = (1 \cdot -1) + (-2 \cdot -2) + (3 \cdot 3)$
 $= -1 + 4 + 9$

$\underline{A} \cdot \underline{B} = 12$

D) $\underline{A} = x - 2y + 3z$
 $\underline{B} = -x - 2y + 3z$

$\underline{A} \times \underline{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & -2 & 3 \\ -1 & -2 & 3 \end{vmatrix}$
 $= \hat{x}((-2 \cdot 3) - (-2 \cdot 3)) - \hat{y}((1 \cdot 3) - (-1 \cdot 3)) + \hat{z}((-2 \cdot 1) - (-2 \cdot -1))$
 $= \hat{x}(-6 - (-6)) - \hat{y}(3 - (-3)) + \hat{z}(-2 - 2)$

$\underline{A} \times \underline{B} = 0\hat{x} - 6\hat{y} - 4\hat{z}$

6) Find the complex phasor vector $\underline{A}$ for the following time-harmonic vectors:

(a) $\underline{A}(t) = \cos(\omega t)\hat{x} - 2\cos(\omega t - \pi/2)\hat{y} - 2\sin(\omega t)\hat{z}$

(b) $\underline{A}(t) = (\sin \omega t - 3\sin \omega t)\hat{y} + (\sin \omega t - \cos \omega t)\hat{z}$

(c) $\underline{A}(t) = 0.3\sin(ky - \omega t)\hat{x}$

A) $\underline{A}(t) = \cos(\omega t)\hat{x} - 2\cos(\omega t - \frac{\pi}{2})\hat{y} - 2\sin(\omega t)\hat{z}$

$A_1 = \cos(\omega t) = 1e^{j0}$

$A_2 = 2\cos(\omega t - \frac{\pi}{2}) = 2e^{-j\frac{\pi}{2}}$

$A_3 = 2\sin(\omega t) = 2\cos(\omega t - \frac{\pi}{2}) = 2e^{-j\frac{\pi}{2}}$

$\underline{A}(t) = e^{j0}\hat{x} - 2e^{-j\frac{\pi}{2}} - 2e^{-j\frac{\pi}{2}}$

B) $\underline{A}(t) = (\sin \omega t - 3\sin \omega t)\hat{y} + (\sin \omega t - \cos \omega t)\hat{z}$

$A_1 = \sin \omega t - 3\sin \omega t$

$A_2 = \sin(\omega t) - \cos(\omega t)$

$A_1 = -2\sin \omega t$

$A_2 = \cos(\omega t - \frac{\pi}{2}) - \cos(\omega t)$

$A_1 = -2\cos(\omega t - \frac{\pi}{2})$

$A_2 = e^{-j\frac{\pi}{2}} - e^j$

$A_1 = -2e^{-j\frac{\pi}{2}}$

$e^j = 1$

$\underline{A}(t) = -2e^{-j\frac{\pi}{2}}\hat{y} + (e^{-j\frac{\pi}{2}} - 1)\hat{z}$

C) $\underline{A}(t) = 0.3\sin(ky - \omega t)\hat{x}$

traveling wave

$\sin(ky - \omega t) = \sin(-(\omega t - ky))$

$= 0.3\sin(-(\omega t - ky))$

$-\frac{\pi}{2}$

$= 0.3\cos(-(\omega t - ky - \frac{\pi}{2}))$

$= 0.3\cos(-\omega t + ky - \frac{\pi}{2})$

make ωt positive

$= 0.3\cos(\omega t - ky + \frac{\pi}{2})$

$\underline{A}(t) = 0.3e^{-j(ky - \frac{\pi}{2})}$

7) From the following complex phasor vectors, find the sinusoidal vector $\underline{K}(t)$ in terms of ωt :

(a) $\underline{K} = j\hat{y} + \hat{z}$

(b) $\underline{K} = j(\hat{y} + j\hat{z})$

(c) $\underline{K} = e^{-jky} \hat{x} - je^{-jky} \hat{z}$

A) $\underline{K} = j\hat{y} + \hat{z}$

$\cos(\omega t + \frac{\pi}{2})$ $\cos(\omega t)$

$-\sin(\omega t)$

$\underline{K}(t) = -\sin(\omega t) \hat{y} + \cos(\omega t) \hat{z}$

B) $\underline{K} = j(\hat{y} + j\hat{z})$

$= j\hat{y} + j^2\hat{z} = -\hat{z}$

$= j\hat{y} - \hat{z}$

$\cos(\omega t + \frac{\pi}{2})$ $\cos(\omega t)$

$= -\sin(\omega t)$

$\underline{K}(t) = -\sin(\omega t) \hat{x} - \cos(\omega t) \hat{z}$

C) $\underline{K} = e^{-jky} \hat{x} - je^{-jky} \hat{z}$

$\underline{K} = \cos(\omega t - ky) \hat{x} - \sin(\omega t - ky) \hat{z}$

8) Sketch (in the xy plane) the trace of the tip of the time-harmonic vector $\underline{A}(t)$, where the corresponding complex phasor vector is

(a) $\underline{A} = 3\hat{x} + 3\hat{y}$

(b) $\underline{A} = 3\hat{x} + j3\hat{y}$

(c) $\underline{A} = 2\hat{x} + j3\hat{y}$

$\cos(\omega t + \frac{\pi}{2}) = -\sin(\omega t)$

A) $\underline{A} = 3\hat{x} + 3\hat{y}$

$3\cos(\omega t)$ $3\cos(\omega t)$

$A(t) = 6\cos(\omega t)$

B) $\underline{A} = 3\hat{x} + j3\hat{y}$

$3\cos(\omega t)$ $3\cos(\omega t + \frac{\pi}{2})$

$= -3\sin(\omega t)$

$A(t) = 3\cos(\omega t) - 3\sin(\omega t)$

C) $\underline{A} = 2\hat{x} + j3\hat{y}$

$2\cos(\omega t)$ $3\cos(\omega t + \frac{\pi}{2})$

$= -3\sin(\omega t)$

$A(t) = 2\cos(\omega t) - 3\sin(\omega t)$

